PROJECT DOCUMENTATION

MarketMind AI

Indian Small Business & Enterprise Sales Intelligence Platform (v2.6)

Infosys Springboard Virtual Internship 7.0

1. Project Information

Project Title: MarketMind AI – Small Business Sales Intelligence Platform (also referred to as "AI Vyapar Copilot"), Version 2.6

Team Members: *Not specified in source material.*

Project Guide:

Not specified in source material.

Repository: Team_1_Small_Biz_Sales_AI

Live Frontend Application: https://marketmind-ai-sales.vercel.app

2. Abstract

MarketMind AI is a full-stack, enterprise-grade AI sales intelligence platform engineered specifically for Indian retail, wholesale, and small-to-medium enterprises (MSMEs). Version 2.6 of the platform helps small businesses, retail stores, supermarkets, and startups monitor sales performance, track inventory, manage invoices, analyze customer behavior, and generate predictive insights. The platform combines a FastAPI backend (SQLAlchemy 2.0, Alembic, Pydantic v2) with a React 18 / Vite frontend (Tailwind CSS, Recharts), and a SQLite (development) / Neon PostgreSQL (cloud production) data layer with a suite of AI/ML models — including multi-horizon time-series sales forecasting (XGBoost, Prophet, Linear Trend), RFM-based customer segmentation and churn risk scoring, a collaborative-filtering and association-rule-mining product recommendation engine, and a multi-factor anomaly detection engine — to support smarter, data-driven business decisions. The system exposes its outputs through role-aware dashboards (Business Owner, Store Manager, Sales Executive, and Platform Administrator workspaces), forecasting reports, customer segmentation views, a bilingual (English/Hindi) AI business copilot, GST-compliant invoicing, and downloadable analytics. The frontend is deployed on Vercel with the database hosted on Neon PostgreSQL. The project outcome is a deployed, role-based, multi-tenant sales intelligence platform that automates data-driven decision-making for small and medium Indian enterprises.

3. Introduction

3.1 Background

The project operates in the domain of retail and small-business analytics, applying Artificial Intelligence and Machine Learning to everyday sales, inventory, and customer data, with a specific focus on the Indian retail and wholesale ("Vyapar") ecosystem. Retail stores, supermarkets, startups, and small enterprises generate continuous streams of sales, inventory, and invoice data but often lack the tools to convert this data into actionable business intelligence.

3.2 Problem Statement

Small businesses need a way to monitor sales performance, track inventory, manage invoices, analyze customer behavior, and generate predictive insights, but typically lack access to integrated, AI-driven sales intelligence tools — including GST-compliant billing and multi-role governance — that automate these data-driven decision-making processes.

3.3 Objectives

- Build a full-stack AI-powered sales intelligence platform tailored to Indian small and medium enterprises.
- Help businesses monitor sales performance, track inventory, manage invoices, and generate GST-compliant billing.
- Analyze customer behavior and generate predictive insights using AI/ML.
- Provide interactive, role-aware dashboards, forecasting reports, customer segmentation, anomaly detection, and intelligent product recommendations.
- Provide a bilingual (English/Hindi) conversational AI business copilot for actionable insights.
- Support smarter business decisions and improve profitability for retail stores, supermarkets, startups, and small enterprises.

4. Scope

4.1 In Scope

- User management with authentication, role-based access control (RBAC), employee invitations, and profile management.
- Mandatory, blocking email verification via 6-digit OTP (Resend API / SMTP dispatch) on first login.
- Sales data upload and management (CSV import, validation, transaction storage).
- Sales and inventory processing (sales aggregation, inventory tracking, revenue calculation, low-stock alerts, supplier purchase order generation).
- Customer segmentation and retention analytics using RFM (Recency, Frequency, Monetary) clustering, with predictive churn risk scoring and Email/WhatsApp outreach workflows.
- AI forecasting engine (multi-horizon revenue and SKU-level demand forecasting for 7, 14, and 30 days).
- Product recommendation engine (collaborative filtering, association rule mining, cross-sell/upsell, customer-specific recommendations).
- Multi-factor anomaly detection module (revenue drops, transaction spikes, inventory discrepancies, irregular discounts) with severity triage and a resolution workflow.
- Analytics dashboard and reporting (role-aware business dashboards, sales/forecast reports, downloadable/CSV analytics).
- 60-day employee task and activity audit trail, filterable by category (Billing & Payments, Inventory & Stock, Logins & Auth).
- Credit receivables aging (0–7, 8–15, 15+ day overdue buckets) with real-time payment visibility.
- GST-compliant invoicing (CGST/SGST/IGST breakdowns; A5 Wholesale, Thermal, and Laser invoice formats).
- Interactive, bilingual (English/Hindi) AI business copilot with domain personas (Executive Strategist, Inventory & Operations Optimizer, Customer Retention Specialist, Data Analyst).
- Platform Administrator console for multi-tenant business directory management and authentication/login audit streaming.

4.2 Out of Scope

Not specified in source material.

5. Existing System & Proposed Solution

5.1 Existing System

Not specified in source material.

5.2 Proposed Solution

MarketMind AI proposes a unified, AI-powered sales intelligence platform that provides role-aware dashboards, multi-horizon forecasting reports, customer segmentation, anomaly detection, GST-compliant invoicing, and an interactive bilingual AI business copilot. The system is intended to help retail stores, supermarkets, startups, and small enterprises automate data-driven decision-making end-to-end, from sales data ingestion and employee activity tracking through to predictive analytics and reporting, within a secure, multi-tenant architecture.

5.3 Key Features

- Role-aware operational workspaces and dashboards for Business Owner, Store Manager, Sales Executive, and Platform Administrator.
- Platform Admin console with dynamic multi-tenant business directory and employee governance.
- Mandatory, blocking email verification via 6-digit OTP.
- Individual sales executive telemetry across configurable time horizons (Today, Yesterday, 7 Days, 30 Days, 6 Months).
- 60-day employee task and activity audit trail.
- Credit receivables aging with 7-day and 15-day buckets.
- Interactive, bilingual AI business copilot with domain-specific personas.
- AI product recommendation engine using collaborative filtering and association rule mining.
- Customer retention and churn analytics with RFM segmentation and outreach workflows.
- Multi-factor anomaly detection with severity triage and a resolution workflow.
- Inventory tracking and automated supplier purchase order generation.
- Multi-model, multi-horizon predictive forecasting (XGBoost, Prophet, Linear Trend) and GST-compliant invoicing.

6. Technology Stack

Category	Technologies
Languages	Python (Backend), JavaScript (Frontend)
Frontend	React 18 (Vite SPA), Tailwind CSS, Vanilla CSS (custom Indian-inspired theme + dark mode), Lucide React icons, Recharts
Backend	Python (FastAPI, asynchronous REST API), SQLAlchemy 2.0, Alembic, Pydantic v2
Database	SQLite (development), Neon PostgreSQL (cloud production)
AI/ML	Scikit-learn, XGBoost, Prophet, Pandas, NumPy
Security	Argon2id (passlib) password hashing, PyJWT (rotating access/refresh tokens), in-memory token-bucket rate limiter
Email / OTP	Resend API, SMTP dispatcher
Testing	Pytest, HTTPX, Pytest-Cov (backend); Vitest, React Testing Library, jsdom (frontend)
Deployment	Vercel (frontend hosting), Neon (cloud PostgreSQL), Git + GitHub

7. System Architecture

High-level data flow: User → Frontend → Backend/API → AI/ML → Database/Storage

The architecture is organized into layers. The User Layer includes Business Owner, Store Manager, Sales Executive, and Platform Administrator roles, each with a dedicated, role-aware workspace. The Data Ingestion & Business Services Layer handles sales data upload, inventory tracking, invoice management (including GST-compliant billing), and dashboard reporting. The API Layer (Gateway & Security) manages authentication (JWT with Argon2id password hashing), mandatory OTP email verification, authorization (RBAC), role and permission management, audit logging/monitoring, and tiered API rate limiting (Authentication: 15 req/min, Public: 60 req/min, Authenticated: 300 req/min). The AI Analytics Engine (ML Models) covers multi-horizon sales forecasting, RFM-based customer segmentation, churn risk scoring, product recommendation, and multi-factor anomaly detection, alongside a non-ML Business Services Layer for upload, invoice, and dashboard services. The Presentation Layer (Dashboard & Reporting) exposes the sales dashboard, inventory overview, customer insights (Customer 360), forecast reports, recommendation panel, anomaly alerts, credit receivables aging, export reports, employee audit trails, and profile settings. The Storage & Database Layer consists of SQLite for local development and Neon PostgreSQL for cloud production, storing transactional, analytics, and user/role data, plus file storage for reports/exports. The Infrastructure & Deployment Layer uses Git, GitHub, and Vercel for frontend hosting, with Neon providing the managed cloud database.

7.1 System Modules

- User Management Module – registration, authentication/authorization, mandatory OTP email verification, role management, employee invitations, profile management.
- Sales Data Upload and Management – CSV upload, validation, transaction storage.
- Sales and Inventory Processing – sales aggregation, inventory tracking, revenue calculation, low-stock alerts, automated supplier purchase order generation.
- Customer Segmentation & Retention Module – RFM (Recency, Frequency, Monetary) clustering, Customer 360 profiles, behavioral analytics.
- AI Forecasting Engine – multi-horizon (7/14/30-day) revenue and SKU-level demand forecasting, seasonal analysis.
- Churn Prediction System – predictive churn risk scoring, retention risk analysis, Email/WhatsApp outreach workflows.
- Product Recommendation Engine – collaborative filtering and association rule mining, cross-sell/upsell suggestions, personalized offers.
- Anomaly Detection Module – multi-factor detection of revenue drops, transaction spikes, inventory discrepancies, and irregular discounts; severity triage and resolution workflow.
- Employee Task & Activity Audit Trail – 60-day chronological audit log of billing, inventory, and login/authentication actions.
- Credit Receivables Aging – 0–7, 8–15, and 15+ day (overdue) buckets with real-time payment visibility.
- AI Business Copilot – bilingual (English/Hindi) conversational assistant with domain personas.
- GST Invoicing Module – GST-compliant A5 Wholesale, Thermal, and Laser invoice generation with CGST/SGST/IGST breakdowns.
- Platform Admin Console – multi-tenant business directory, authentication/login audit stream, AI model retrain schedules.
- Analytics Dashboard and Reporting – role-aware business dashboards, sales/forecast reports, downloadable analytics.

8. AI/ML Implementation

8.1 Dataset

Retail sales and inventory datasets are collected and analyzed for structure and business attributes as part of the initial project milestone. Sample datasets and generated model artifacts are maintained in the repository's data/ directory, with ML training pipelines, data cleaning, and feature engineering handled in the preprocessing/ directory. Specific dataset source, size, and format are not detailed in the source material.

8.2 Data Processing

- Sales data upload via CSV import.
- Data validation and transaction storage (implemented with FastAPI, SQLAlchemy 2.0, and PostgreSQL/SQLite).
- Cleaning and preprocessing of transaction records via dedicated ML training pipelines (preprocessing/ directory).
- Processing of inventory and customer datasets.

8.3 Model / Algorithm

Module	Model Type	Algorithms	Key Inputs	Key Outputs
Sales Forecasting Engine	Time Series Forecasting (multi-horizon: 7/14/30 days)	XGBoost, Prophet, Linear Trend	Historical sales, product sales, seasonal trends, revenue history	Revenue forecast, SKU-level demand forecast, trend analysis
Customer Segmentation & Retention	Unsupervised / Behavioral Clustering	RFM (Recency, Frequency, Monetary) Segmentation	Purchase recency, frequency, purchase value, customer activity	Customer tiers (Champions, Loyal, At Risk, Hibernating), behavioral segments
Churn Prediction System	Predictive Risk Scoring	Predictive churn risk scoring (specific algorithm not detailed in source material)	Purchase inactivity, order frequency, customer engagement	Churn risk score, revenue-at-risk customers
Product Recommendation Engine	Recommendation System	Collaborative Filtering, Association Rule Mining	Purchase history, product combinations, customer preferences	Personalized recommendations, cross-sell/upsell bundles, estimated revenue uplift
Anomaly Detection Module	Multi-Factor Anomaly Detection	Isolation Forest, Statistical Outlier Detection	Sales transactions, inventory changes, revenue patterns, discount patterns	Severity-tagged alerts (Critical, Warning, Info): revenue drops, transaction spikes, inventory discrepancies

8.4 Training & Evaluation

Each AI/ML module is evaluated using the following metrics:

- Sales Forecasting: MAE, RMSE, R^2 Score, with chronological train/validation splits and baseline improvement gates.
- Customer Segmentation: Silhouette Score.
- Churn Prediction: Accuracy, Precision, Recall, F1-Score.
- Recommendation System: Precision@K, Recall@K.
- Anomaly Detection: Detection Accuracy, False Positive Rate.

8.5 AI/ML Workflow

Input → Processing → Model → Output

For example, in the Sales Forecasting Engine: historical sales, product sales, seasonal trends, and revenue history (Input) are processed and fed into XGBoost / Prophet / Linear Trend models (Model), producing multi-horizon (7/14/30-day) revenue forecasts, SKU-level demand predictions, and trend analysis (Output), with model health tracked via chronological train/validation splits and MAE/RMSE/R^2 metrics.

9. Database & API

9.1 Database

The storage layer uses SQLite for local development and Neon PostgreSQL for cloud production, and consists of the following entities:

- Transactional data – sales, inventory, and invoice transaction records.
- Analytics data – aggregated data supporting reports and exports.
- User & Role data – user accounts, roles, and role-based access control data, managed via SQLAlchemy 2.0 models and Alembic migrations.
- File Storage – reports and exports.

Schema/ER diagram:

Not specified in source material.

10. User Interface

The platform features custom, Indian retail ("Vyapar")-inspired branding: a Dukaan awning-and-arch silhouette logo symbolizing traditional storefronts, an integrated Indian Rupee (₹) symbol, an ascending sales-momentum wave motif representing growth, and a warm Saffron, Gold, and Royal Indigo color palette. The frontend supports light and dark themes and bilingual (English/Hindi) display. Specific UI mockups or screenshots are not included in the source material.

11. GitHub & Version Control

11.1 Repository

GitHub Repository: https://github.com/springboardmentor24052s-tech/Team_1_Small_Biz_Sales_AI

Main Branch:

Not specified in source material.

11.2 Repository Structure

- backend/ – alembic/ (database migrations), app/api/v1/ (REST endpoints: auth, users, audit, sales, inventory, team, forecasts), app/core/ (security, rate limiting, JWT, config, CORS), app/db/, app/models/, app/schemas/, app/services/ (business logic, ML models, email dispatchers), app/commands/ (CLI seeding), bootstrap.py, main.py; tests/ (Pytest suite, 48 tests); requirements.txt, pyproject.toml.
- frontend/ – src/components/ (common, dashboards, modules), src/context/ (Auth, Language, Toast, Theme), src/services/ (Axios API clients with auto token refresh), src/test/ (Vitest & React Testing Library, 31 tests); public/ (static assets, favicon); package.json, vite.config.js.
- preprocessing/ – ML training pipelines, data cleaning, and feature engineering.

- data/ – sample datasets and generated model artifacts.
- docs/ – architectural documentation and project blueprints.

11.3 Version Control

Version control is managed using Git and GitHub. Further detail on branching, commits, pull requests, and issue tracking is not specified in the source material.

12. Error Handling & Security

12.1 Error Handling

The platform uses zero-leakage error handling: sanitized global exception handlers for database errors (SQLAlchemyError) and internal runtime errors, with server-side correlation IDs to prevent internal schema leakage to clients.

12.2 Security

- Authentication and authorization via JWT, with rotating access and refresh token sessions.
- Argon2id (passlib) password hashing.
- Role-Based Access Control (RBAC) with defined permissions and restrictions per role (Business Owner, Store Manager, Sales Executive, Platform Administrator).
- Mandatory, blocking 6-digit OTP email verification (Resend API / SMTP) for first-time dashboard access, registration, and password resets.
- Tiered, in-memory token-bucket rate limiting: Authentication (15 req/min), Public (60 req/min), Authenticated (300 req/min), with automated Retry-After headers.
- Tenant data isolation – strict multi-tenant boundaries; new registrations start with clean, private workspaces, and sample data is isolated to demo accounts.
- No hardcoded secrets – credentials, JWT secrets, database connection strings, and API keys are configured via environment variables with typed Pydantic validation.
- File upload protection – avatar image uploads verified via magic-byte file signature validation (image/png, image/jpeg, image/webp) with a 2MB size cap.
- Audit logging & monitoring – comprehensive authentication and login audit stream (auth_logs) and a 60-day employee task/activity audit trail.
- API routing & rate limiting at the gateway layer.

13. Testing

MarketMind maintains an extensive, fully automated test suite across backend and frontend stacks:

Test Category	Suite Coverage	Passed	Failed	Pass Rate
Backend API & Core	Auth, RBAC, Rate Limiting, File Safety, Forecasting, Recommendations, Anomaly, Churn, Team	48	0	100%
Frontend UI & Services	Auth Context, Token Refresh, Error States, Recommender, Anomaly Actions, Forecasting Views, Admin Governance	31	0	100%
Total Automated Tests	End-to-end integration and unit verification	79	0	100%

- Backend tests executed via Pytest (pytest -v); Frontend tests executed via Vitest (npm run test:run).

- Frontend code quality: 0 ESLint errors, 0 npm audit vulnerabilities.
- Production build: bundle compilation succeeds cleanly with npm run build.
- Database integrity: clean Alembic migration schemas, verified compatible with SQLite (local) and Neon PostgreSQL (cloud).

14. Results

14.1 AI/ML Results

The platform reports the following metric categories per module (specific numeric results are not provided in the source material):

- Sales Forecasting: MAE, RMSE, R^2 Score, with baseline improvement gates.
- Customer Segmentation: Silhouette Score.
- Churn Prediction: Accuracy, Precision, Recall, F1-Score.
- Recommendation System: Precision@K, Recall@K.
- Anomaly Detection: Detection Accuracy, False Positive Rate.

14.2 System Results

The automated test suite reports 79 of 79 tests passing (100% pass rate) across backend (48 tests) and frontend (31 tests), with 0 ESLint errors and 0 npm audit vulnerabilities. Further system-level results (e.g., performance benchmarks, user acceptance results) are not specified in the source material.

15. Deployment

The frontend application is deployed on Vercel, with the cloud database hosted on Neon PostgreSQL; local development uses SQLite. Source control and collaboration are managed via Git and GitHub.

16. Challenges & Limitations

Not specified in source material.

17. Future Scope

Not specified in source material.

18. Conclusion

MarketMind AI addresses the need for an integrated, AI-powered sales intelligence platform for Indian small businesses by combining sales, inventory, and GST-compliant invoice management with AI/ML-driven multi-horizon forecasting, RFM-based customer segmentation, churn risk scoring, product recommendations, and multi-factor anomaly detection. The platform adds role-aware workspaces, a 60-day employee audit trail, credit receivables aging, mandatory OTP-secured authentication, and a bilingual (English/Hindi) AI business copilot within a secure, multi-tenant architecture. Deployed with a React/Vite frontend on Vercel and a FastAPI backend backed by Neon PostgreSQL, and validated by a fully automated, 100%-passing test suite (79 tests), the project results in a deployed, role-based platform delivering interactive dashboards and AI-generated business insights to support smarter, data-driven decision-making and improved profitability for retail stores, supermarkets, startups, and small enterprises.

19. References

Repository: https://github.com/springboardmentor24052s-tech/Team_1_Small_Biz_Sales_AI
Live Application: https://marketmind-ai-sales.vercel.app